

DATA PARSING OPTIONS			
Column separator	\t Tab		
Thousands separator	,		
Decimals separator	.		
Date Locale	en-GB		
DATA TRANSFORMATION			
Stack on	Column		
Reset			

	Aa Letter	Aa Language	# Frequency (%)
105	a	Esperanto	12.12
106	i	Esperanto	10.01
107	e	Esperanto	9
108	o	Esperanto	8.78
109	n	Esperanto	7.96
110	l	Esperanto	6.1
111	s	Esperanto	6.09

Figure 4: Sample data from a repository

Alluvial Diagram
It shows correlations between categorical dimensions representing them as flows, visually linking categories with shared items. Each rectangle represents a unique value in the selected dimension, its height is proportional to its value. Correlations are represented with curved lines whose width is proportional to their value.

[Code](#) [Tutorial](#)

Alluvial Diagram
Correlations, proportions

Arc Diagram
Networks

Bar chart
Correlations

Multi-set bar chart
Correlations, proportions

Stacked bar chart
Correlations, proportions

Beeswarm plot
Distributions, time series, proportions

Box plot
Distributions

Bubble chart
Correlations, proportions

Bumpchart
Time series, correlations, proportions

Calendar heatmap
Time chunks, proportions

Circle Packing
Hierarchies, proportions

Circular dendrogram
Hierarchies, proportions

Contour plot
Correlations, distributions

Convex hull
Correlations, proportions

Linear dendrogram
Hierarchies, proportions

Gantt chart
Time series, correlations

Hexagonal binning
Correlations, distributions

Horizon graph
Time series, correlations

Figure 5: Chart types in RAWGraphs

pipeline (Figure 2) of RAWGraphs are:

- Data loading
- Chart selection
- Data dimension mapping to the visual variables of the chart
- Customisation
- Exporting

Data loading

The RAWGraphs web app can be accessed at <https://app.rawgraphs.io/>. The first step in the visualisation pipeline is to load the data. The following options are provided in this step:

- Paste your data
- Upload your data

- Try data samples
- SPARQL query
- From URL
- Open your project

When you select the option 'Upload your data', RAWGraphs allows you to choose from the following file formats:

- .tsv (Tab separated values)
- .csv (Comma separated values)
- .dsv (Delimiter separated values)
- .json (JavaScript object notation)

In the case of .json files, the level of the nested structure can be selected to transform the data into a table.

Figure 4 shows sample data loaded from the repository 'Letter frequency by

language'. This repository has 390 rows (1170 cells).

Chart selection

The second step is to select the chart type. The power of RAWGraphs lies in the variety and specific types of charts it provides. Figure 5 illustrates a set of charts this tool can render.

The best part is that, for each chart type, a clear description along with a demo video is provided, helping users make an informed choice.

The most important task in the data visualisation process is the selection of suitable types of visualisation. Incorrect